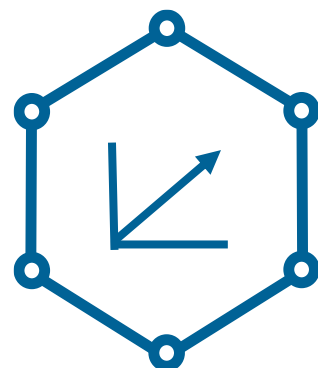


Knowledge Graphs

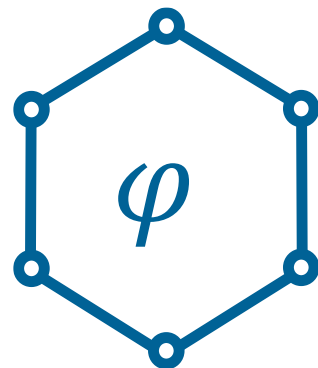


Emanuel Sallinger



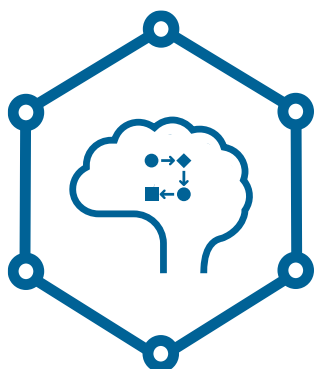
KG Embeddings

Widely-applied, large family of ML models



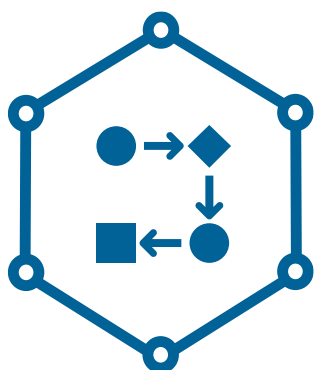
Logical Knowledge in KGs

Highly expressive, diverse family of logical models.



Graph Neural Networks

Using the KG structure *as* a neural network.



Data Models

Overview of data models in different communities



Graph Neural Networks



Message Passing Neural Networks



GraphSAGE



Graph Convolutional Networks



Applications



Theory to Practice